

An Editable Browser for Reranking Web Search Results

Takehiro Yamamoto¹

Satoshi Nakamura¹

Katsumi Tanaka¹

¹*Department of Social Informatics, Graduate School of Informatics,
Kyoto University Yoshida-Honmachi, Sakyo, Kyoto 606-8501 Japan
{tyamamot, nakamura, tanaka}@dl.kuis.kyoto-u.ac.jp*

Abstract

This paper proposes methods of reranking Web search results according to editing operations the user performs while browsing the search results. Users' Web search intentions are diverse. As a result, it is difficult for search engines to return good search results which satisfy the user's search intention adequately. Hence, the user must check search results sequentially, or re-search creating a new query. Our goal is to reflect the users' search intentions by editing search results through user interaction and propagating the intention of the editing operations. In this paper, we propose methods of reranking Web search results according to the user's deletion and emphasis operations. We describe the implementations of these reranking methods and provide evaluations.

1. Introduction

Recently, the performance of Web search engines has increased amazingly because the number of Web pages is increasing explosively. A variety of ranking algorithms have been proposed and used in many Web search engines. People search for information which they need using search engines. Search engines, however, cannot always return good search results which satisfy users' search intentions adequately.

For example, if a product called "A" is used as a query, one user might be seeking reviews of "A", while another user might be seeking how to buy "A". Thus, although users often use the same query in a Web search, their search intentions may differ from other users. Moreover, search intentions might be diverse even if the same user uses the same query. Hence, it is difficult to recognize users' search intentions just by analyzing their input queries.

Because of this diversity, the search results which a user needs do not correspond to his search intentions. In this case, the user must check the search results sequentially until he obtains sufficient information from the linked pages from the page containing the search results. It is tiring for users to check a large number of search results, so many users give up searching after

checking a few high ranked results. If the user cannot find adequate target information, he needs to re-search by creating a new query. However, it is difficult to create a new query which refines the search results effectively. In particular, when the user cannot verbalize what he wants to search for, it is too difficult to recreate a query.

As we mentioned above, when the search results that the user needs are not ranked highly, or he predicted to these results ranked lowly, the user wants to rerank the search results by degrading the high ranked irrelevant results or upgrading the low ranked results that he/she wants. However, conventional search engines do not support this kind of reranking. One reason for this is the limited interaction between user and search engine. In conventional search engines, user's operations are limited only to inputting keywords, scrolling the page of search results, and clicking URLs to visit.

Thus, the search engines cannot reflect the user's search intentions such as "I don't need these results about this topic", or "I want to raise the ranking of a certain results page about this topic."

By the way, we use many kinds of operations in daily applications. For example, when we edit a document, we delete improper text, or emphasis important text. We often drag-and-drop files to a trash can. These operations are very useful, but there are no such functions in the page of Web search results.

In this work, we introduce editing operations such as *deletion* and *emphasis* that can be employed while users are browsing Web search results. Our system enables users to edit any portion of the page of Web search results at any time of while searching. Our system detects users' search intentions from the editing operation. Then our system propagates users' search intentions based on their editing operation to all of the Web search results.

Our system guesses the user's search intentions, for example assuming, "This user does not want this kind of the result", if the user deletes some areas of the search results, or "This user wants this kind of the results more", if the user emphasizes some area of the search results. After estimating the user's search intentions, our system reranks the search results according to the intention and shows the reranked results to the user. In this way, the user can easily obtain optimized search results.

The organization of this paper is as follows. We discuss related work in the next section. In section 3, we describe the user's operations and the search intentions. In section 4, we propose reranking methods when a user performs deletion or emphasis operations. In section 5, we describe the implementation of our system. Section 6 and section 7 explain our evaluation experiments, and discuss their results. Finally, we conclude this paper and describe the future work.

2. Related work

2.1 Web browsing and interactions

There are many studies which have focused on novel interaction techniques for Web browsing. *Zero-Click*[7] and zooming cross-media[1] allow users to browse Web content without clicking. *Zero-Click* allows users to browse linked Web content by simply locating the mouse cursor in the target content. However, these studies have only focused on how easy to browse the linked Web contents. They do not consider reranking the search results.

2.2 Search intention

There are many studies related to the user's search intention. Broder[2], Rose and Levinson[9] classified the goal of a user into three categories. There are many studies on automatic classification of user goals by analyzing queries[5][6]. Tian et al.[12] classified relations of multiple keywords into two categories and proposed a method of improving Web retrieval precision by using proximity and density of multiple keywords. These works focused on recognizing users' search intentions just by analyzing their queries. However, our work is to guess the user's search intention through interactions between the user and the system, and to reflect this intention in the Web search results page.

Yahoo! Mindset[13] allows the user to specify his search intentions to the system. *Yahoo! Mindset* weights each Web page as 'more research' or 'more shopping'. The user can specify that his search intention is research or shopping by using a slide bar; then the system re-ranks the search results according to the weight. However, in this system, the user cannot do reranking according to the user's search intention without prepared factors.

2.3 Reranking method

Relevance feedback[11] is the most popular retrieval system for retrieving documents using the user's feedback. Relevance feedback is based on vector space models[12]. Non-relevance feedback[9] is a retrieval method for

documents. This uses only non-relevant documents to find the target documents from a large data set of documents. We think that we can apply relevance feedback and non-relevance feedback to our system.

3. Reflection of the user's search intentions using editing operations

3.1 Interaction between users and the system

Our system reranks the search results according to the user's editing operations and the target that the user edits. The processing flowchart is shown in Figure 1.

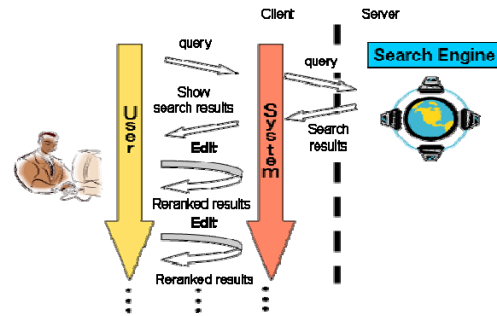


Figure 1. System image

1. The user inputs a query to our system.
2. The system sends the query to a Web search engine.
3. The system receives the search results and shows them to the user.
4. The user checks the search results received from the system.
5. The user edits any portion of the search results by using editing operations,.
6. The system detects the user's editing operations and guesses the user's search intention. Then the system reranks the search results according to this guess and shows the new reranked search results to the user.
7. If the user is not satisfied with the reranked search results, then the system returns to step 4.

3.2 Type of editing operations

Deletion operations

When editing documents, we use deletion operations to delete wrong or unnecessary text. If users could use deletion operations in browsing the Web search results, they might want to delete some keywords or search results which were obviously irrelevant to the results they need. Therefore, in this paper, we consider a deletion operation in which a user can send the system information on what types of search results are not wanted.

Emphasis operation

We use emphasis operations to highlight text and therefore these operations are exactly opposite to deletion operations. Thus, in this paper, we consider an emphasis operation in which a user can send information on what types of search results are wanted.

3.3 Target of the operation and a user's search intention

Search engines usually return a Web search results page like Figure 2 to the user.

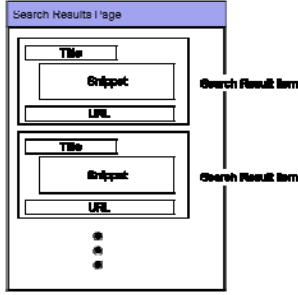


Figure 2. An example of a search results page

Title

The title of a Web page usually consists of a few keywords. If a user deletes a keyword from the title of a search result, we can guess that he wants to degrade results which include the deleted keyword. We can also guess that he wants to degrade results which include topics on the deleted keyword. For example, while browsing the Web search results of “Katsumi Tanaka”, if a user deletes “pianist” from the title of a search result, we can guess that he has the intention to degrade search results related to the topic of Katsumi Tanaka, who is a pianist.

Snippet

A snippet consists of a few sentences that form a summary of a Web page according to a query. The user might delete not only the keywords of a snippet but also the sentences of the snippet. If a user deletes a sentence of a snippet, we can guess that he wants to degrade the search results which include the topic related to the deleted sentences. For example, if a user wants to degrade results which describe an opinion such as agreement, he will delete a sentence like “I agree on --.”

URL

If a user deletes a URL or a part of a URL, we can guess that he wants to degrade the results which include the deleted strings. If results containing the

domain identifier JP are unnecessary for the user, he might delete “.jp” from a URL of the search results.

Search result item

A search result item, which represents a search result, consists of meta-data such as its title, snippet, and URL. If a user deletes a search result item, we can guess that he wants to degrade the search results which are similar to the deleted search result item.

Multiple search result items

If the first search result item is relevant and the second and third search result items are irrelevant to the user's target content, the user might delete the second and third results at the same time. In this case, we can reflect the user's search intention by upgrading the search results which are similar to the first result and degrading the search results which are similar to the second and third results.

In any case, if a user uses emphasis operations instead of deletion operations, we can guess he wants to upgrade the search results which he/she is looking for. The intentions which we can guess from the user's emphasis operations are exactly opposite to those of deletion operations.

4. Reranking methods

Our goal is to recognize a user's search intention from his editing operations and to rerank the search results according to the recognized search intention. For this purpose, we propose the reranking of Web search results by deleting or emphasizing a keyword. The reranking process is as follows;

1. The user inputs a query q to our system.
2. The system sends q to a Web search engine, and then the system obtains the top N search results:
 $R = \{r_1, r_2, \dots, r_N\}$ (r_i is the i^{th} search result)
3. For each search result item r_i , the system calculates the score of r_i according to the following formula:
$$Score(r_i) = N - i \quad (1)$$
4. The system records the search result scores and renders them for the user. Then, the system awaits the user's operations.
5. When the system detects the user's editing operations, it calculates the score of all search results by using the prepared algorithms. Then the process jumps to step 3.

4.1 Selection keyword method

When the user deletes or emphasizes the keyword t , this method degrades or upgrades the search results which include t . Our system calculates the score of r_i which includes t in its title or snippet according to Equation (2) when the user deletes, or Equation (3) when the user emphasizes.

$$Score(r_i)_{new} = Score(r_i)_{last} - N \quad (2)$$

$$Score(r_i)_{new} = Score(r_i)_{last} + N \quad (3)$$

4.2 Extended keyword method

For example, if a user starts searching by inputting a query like “Katsumi Tanaka”, and then deletes the keyword “pianist” in the search result, the system guesses the user’s search intention to be “the user wants to degrade the results about ‘Katsumi Tanaka who is a pianist’”. So when the user deletes “pianist”, it is better for the user to degrade the search results which not only include “pianist” but also include “piano” or “concert” and so on. In order to accomplish this purpose, we propose a method called the “extended keyword method” using the difference of the term frequency. The extended keyword method uses the fact that the terms like “concert” or “piano” appear more frequently in the search results of “Katsumi Tanaka AND pianist” than those of “Katsumi Tanaka”. The algorithm to obtain the outer extended keywords is as follows:

1. For pre-processing, the system performs morphological analyses about the search result page p of query q , and calculates the set of keywords W appear in p .
2. For each keyword $w \in W$, the system calculates the term frequency $TF(w, p)$ in R .
3. The system obtains the keyword t when the system detects a user’s deletion or emphasis operation.
4. The system sends the query “ q AND t ” to a Web search engine, and obtains the top N search results. (We call this search results page p').
5. For each keyword $w \in W$, the system calculates the term frequency $TF(w, p')$ in the search results page p' .
6. The system extracts the top k terms in order of the values of $TF(w, p') - TF(w, p)$.
7. For each keyword t , $w_1, w_2, \dots, w_k$, the system recalculates the score of the search result items

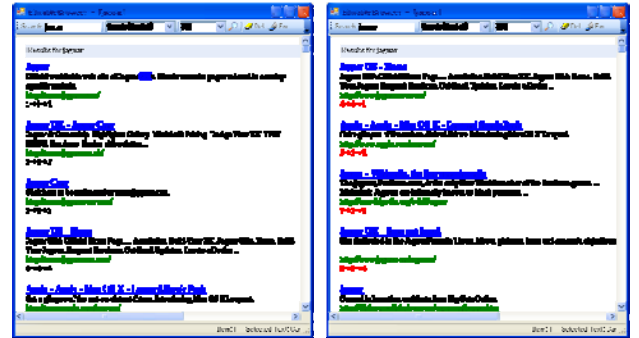
$r_i \in R$ which include these keywords according to Equation (2) if the user uses the deletion operation, or Equation (3) if the user uses the emphasis operation.

For example, by using this method, when the user deletes “pianist” in the search results of “Katsumi Tanaka”, the extended keyword method degrades the search results which include “profile”, “concert”, “lesson”, or “piano”. (If the system uses Google[4] and sets ($N = 500$ and $k = 4$).

5. Implementation

In order to realize our reranking methods, we implemented a prototype system. In this work, by using Japanese morphological analyzer ChaSen[3], we extract proper nouns and general nouns as keywords.

An example of implementation of our prototype is shown in Figure 4. First, users search with specifying what search engines the system uses and the number of search results which the system obtains. Then the user can delete or emphasize a keyword by selecting the keyword and pushing the delete button or emphasis button placed at the top of the system. The user then receives the reranked search results.



(a) before (b) after
Figure 4. Examples of our system: The search results are reranked after deleting a keyword

6. Experimentation

In order to evaluate our proposed reranking methods, we experimented on how the search results are reranked by deleting or emphasizing keywords using the reranking methods proposed in section 4. We prepared 15 search tasks and we also prepared correct answer sets for each search task.

6.1 Keyword deletion task

In this task, the user inputs the prepared search query and submits the query to the search engine using our system, and then receives the search results. Then, the user checks the search results sequentially. If the user obtains an unmatched search result, he selects the keyword and deletes it. If the system detects the user's deletion, the system reranks the search results and displays the reranked search results. Then the user also rechecks the search results from top to bottom. The user does this three times. After each deletion, we calculated the precision of the ranking and the number of correct search results which the system unfortunately degraded. We compared the selection keyword method (SKM) and the extended keyword method (EKM) in this task.

6.2 Keyword emphasis task

In this task, the user inputs the search query and submits the search query to the search engine using our system, and then receives the search results. Then, the user checks the search results sequentially. If the user finds a keyword which he thinks is relevant, he selects the keyword and emphasizes it. If the system detects the user's emphasis operation, the system reranks the search results and displays reranked search results. Then the user also rechecks the search results from top to bottom. The user does this three times. After each emphasis, we calculated the precision of the ranking. We compared selection the keyword method and the extended keyword method in this task.

In both tasks, if all reranked top 20 search results are correct, the user stops the operation.

6.2 Experimental results

Figure 5 shows the average precision of the top K ($K = 1, 2, \dots, 20$) search results of each deletion. Figure 6 shows those of each emphasis. Table 1 shows the numbers of the matched search results which are removed from the top 20 by the deletion. The number of total deletions means the total number of search results which were removed from the top 20 by three deletions. The number of error deletions means the number of matched search results which were irrelevantly removed from the top 20 by three deletions. Error ratio means the ratio of the total deletion number and the error number.

7. Discussions

7.1 Keyword deletion task

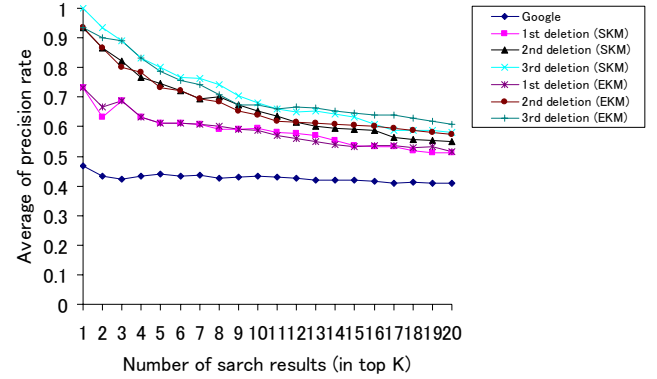


Figure 5. Average precision of each deletion

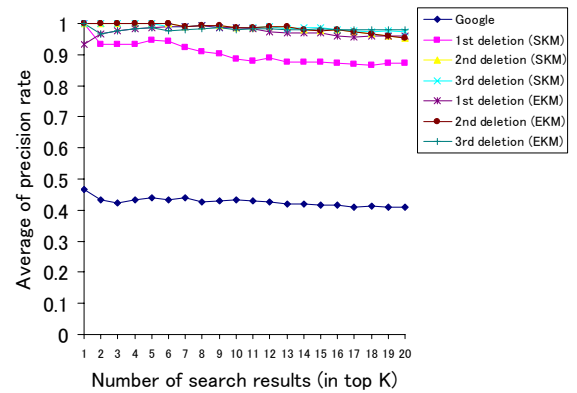


Figure 6. Average precision of each emphasis

Table 1. Ratio of matched search results which were removed from the top 20

	Number of total deletions	Number of error deletions	Error ratio
SKM	127	2	1.57%
EKM	191	20	10.47%

Figure 5 shows that the number of relevant search results in the top 20 search results increased after each deletion operation in both reranking methods. We note that the effect of the first deletion was greatest and the effect of the deletion became smaller as the user repeated the deletion. In the meantime, the difference between the effectiveness of the extended keyword method and that of the selection keyword method was relatively small.

In particular, if we assume that users usually check only the top 10 search results, the effectiveness of both reranking methods is the same. As shown in Table 1, the main reason for this fact is that although the extended keyword method can degrade more search results than the selection keyword method, the extended keyword method also degrades more relevant search results. This problem occurs because the extended keyword method obtains so many “popular terms” or “terms which appear in multiple

topics” that many relevant search results are also degraded. In order to increase the accuracy of the extended keyword method, we need to resolve this problem.

7.2 Keyword emphasis task

The effect of emphasis operations is much greater than that of delete operations. We note that the first emphasis operation is effective, since all the top 20 search results were relevant in more than 2/3 queries.

In the keyword emphasis task, the extended keyword method outperformed the selection keyword method. In the keyword deletion task, the fact that the extended keyword method often obtained popular terms like “subway”, which are related to multiple topics, caused a decrease in accuracy. In the emphasis operation task, however, even if the extended keyword method obtains such popular keywords, if the keyword which the user emphasized directly or other keywords which the extended keyword method obtains are related to relevant search results, the decrease in precision will not occur. We think the extended keyword method outperformed the selection keyword method for this reason.

7.3 Deletion operations and emphasis operations

It is more difficult for users to judge what terms are related to their information needs than to judge what terms are *not* related. Therefore, if we use emphasis operations to rerank search results, we need to verbalize what we want to search for in concrete form.

In this work, we used queries for which collection sets were easy to prepare, so we were able to find terms which we should emphasize. However, we think that users often cannot find terms to be emphasized in a real search. Users may gradually optimize search results by combining deletion and emphasis operations.

9. Conclusion and future work

In this paper, we proposed a technique to guess users’ search intentions by allowing users to edit search results and to reflect these intentions in search results. Then, we proposed reranking methods involving the user deleting or emphasizing keywords, and we evaluated these methods. We showed the difference between the effectiveness of deletion operations and emphasis operations.

In the future, we will work on not only deletion or emphasis of keywords but also of search result items or sentences in a snippet. In addition, we will work on assuming and reflecting users’ search intentions using other operations such as drag and drop.

The technique which we proposed in this paper can be applied to other types of browsing. For example, while browsing a bulletin board, users can delete improper messages or spam to enhance readability, or emphasize a term so that they read only topics which they need. We will develop an editable browser for many kinds of Web pages.

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